VICTORIAN CURRICULUM AND ASSESSMENT AUTHORITY

Victorian Certificate of Education
2019

# PHYSICS

## Written examination

### FORMULA SHEET

**Instructions**

This formula sheet is provided for your reference.
A question and answer book is provided with this formula sheet.

**Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.**

# Physics formulas

## Motion and related energy transformations

| | |
|---|---|
| velocity; acceleration | $v = \frac{\Delta s}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$ |
| equations for constant acceleration | $v = u + at$<br>$s = ut + \frac{1}{2}at^2$<br>$s = vt - \frac{1}{2}at^2$<br>$v^2 = u^2 + 2as$<br>$s = \frac{1}{2}(v + u)t$ |
| Newton's second law | $\Sigma F = ma$ |
| circular motion | $a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$ |
| Hooke's law | $F = -k\Delta x$ |
| elastic potential energy | $\frac{1}{2}k(\Delta x)^2$ |
| gravitational potential energy near the surface of Earth | $mg\Delta h$ |
| kinetic energy | $\frac{1}{2}mv^2$ |
| Newton's law of universal gravitation | $F = G\frac{M_1 M_2}{r^2}$ |
| gravitational field | $g = G\frac{M}{r^2}$ |
| impulse | $F\Delta t$ |
| momentum | $mv$ |
| Lorentz factor | $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$ |
| time dilation | $t = t_o\gamma$ |
| length contraction | $L = \frac{L_o}{\gamma}$ |
| rest energy | $E_{rest} = mc^2$ |
| relativistic total energy | $E_{total} = \gamma mc^2$ |
| relativistic kinetic energy | $E_k = (\gamma - 1)mc^2$ |

## Fields and application of field concepts

| | |
|---|---|
| electric field between charged plates | $E = \frac{V}{d}$ |
| energy transformations of charges in an electric field | $\frac{1}{2}mv^2 = qV$ |
| field of a point charge | $E = \frac{kq}{r^2}$ |
| force on an electric charge | $F = qE$ |
| Coulomb's law | $F = \frac{kq_1q_2}{r^2}$ |
| magnetic force on a moving charge | $F = qvB$ |
| magnetic force on a current carrying conductor | $F = nIlB$ |
| radius of a charged particle in a magnetic field | $r = \frac{mv}{qB}$ |

## Generation and transmission of electricity

| | |
|---|---|
| voltage; power | $V = RI$; $P = VI = I^2R$ |
| resistors in series | $R_T = R_1 + R_2$ |
| resistors in parallel | $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ |
| ideal transformer action | $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$ |
| AC voltage and current | $V_{RMS} = \frac{1}{\sqrt{2}}V_{peak}$ $I_{RMS} = \frac{1}{\sqrt{2}}I_{peak}$ |
| electromagnetic induction | EMF: $\varepsilon = -N\frac{\Delta\Phi_B}{\Delta t}$ flux: $\Phi_B = B_\perp A$ |
| transmission losses | $V_{drop} = I_{line}\,R_{line}$ $P_{loss} = I^2_{line}\,R_{line}$ |

## Wave concepts

| | |
|---|---|
| wave equation | $v = f\lambda$ |
| constructive interference | path difference = $n\lambda$ |
| destructive interference | path difference = $\left(n - \frac{1}{2}\right)\lambda$ |
| fringe spacing | $\Delta x = \frac{\lambda L}{d}$ |
| Snell's law | $n_1 \sin\theta_1 = n_2 \sin\theta_2$ |
| refractive index and wave speed | $n_1 v_1 = n_2 v_2$ |

## The nature of light and matter

| | |
|---|---|
| photoelectric effect | $E_{\text{k max}} = hf - \phi$ |
| photon energy | $E = hf$ |
| photon momentum | $p = \dfrac{h}{\lambda}$ |
| de Broglie wavelength | $\lambda = \dfrac{h}{p}$ |

## Data

| | |
|---|---|
| acceleration due to gravity at Earth's surface | $g = 9.8\ \text{m s}^{-2}$ |
| mass of the electron | $m_e = 9.1 \times 10^{-31}\ \text{kg}$ |
| magnitude of the charge of the electron | $e = 1.6 \times 10^{-19}\ \text{C}$ |
| Planck's constant | $h = 6.63 \times 10^{-34}\ \text{J s}$ $h = 4.14 \times 10^{-15}\ \text{eV s}$ |
| speed of light in a vacuum | $c = 3.0 \times 10^{8}\ \text{m s}^{-1}$ |
| gravitational constant | $G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$ |
| mass of Earth | $M_E = 5.98 \times 10^{24}\ \text{kg}$ |
| radius of Earth | $R_E = 6.37 \times 10^{6}\ \text{m}$ |
| Coulomb constant | $k = 8.99 \times 10^{9}\ \text{N m}^2\ \text{C}^{-2}$ |

## Prefixes/Units

| | | | |
|---|---|---|---|
| p = pico = $10^{-12}$ | n = nano = $10^{-9}$ | $\mu$ = micro = $10^{-6}$ | m = milli = $10^{-3}$ |
| k = kilo = $10^{3}$ | M = mega = $10^{6}$ | G = giga = $10^{9}$ | t = tonne = $10^{3}$ kg |

**END OF FORMULA SHEET**